Learning Camera Geometry from Unlabeled Real-World Dynamic Video

Master's Thesis Defense

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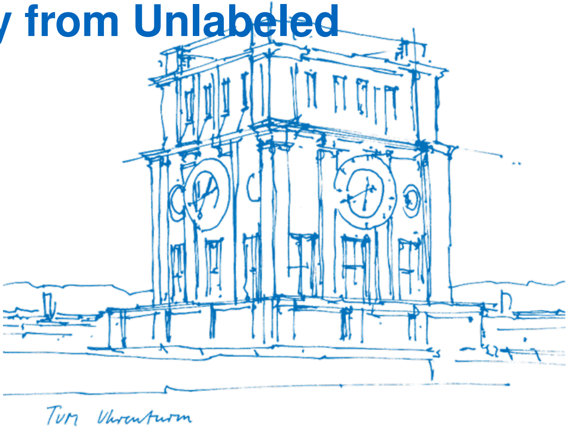

Outline

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Motivation

Recover **camera poses** (R, t) and **calibration** $(\mathbf{K})$ from **unlabeled casual video**

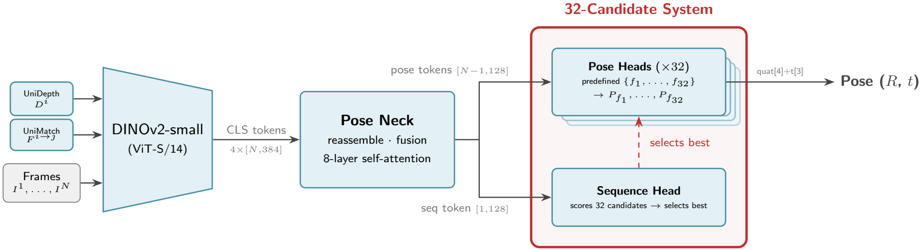
Why is this hard?

- Casual video: dynamic objects, unknown cameras, no 3D labels
- Classical SfM/SLAM fails on dynamic scenes
- Supervised methods need expensive ground truth that doesn't scale

What exists today:

- Strong pretrained models for individual sub-tasks: depth, optical flow, calibration, pose
- But they operate **in isolation** — no joint reasoning, no temporal consistency

The Calibration Problem in AnyCam



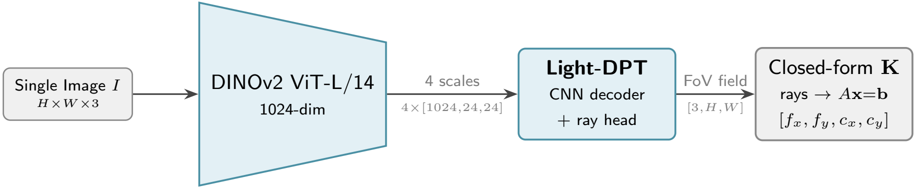
32-candidate focal length system:

- Generates 32 focal length guesses
- Runs flow reprojection for each
- Sequence head selects the best

AnyCam calibration error:

Dataset	Focal length MAPE
Sintel	**45.1%**
TUM-RGBD	**63.7%**

AnyCalib — A Path to Better Calibration

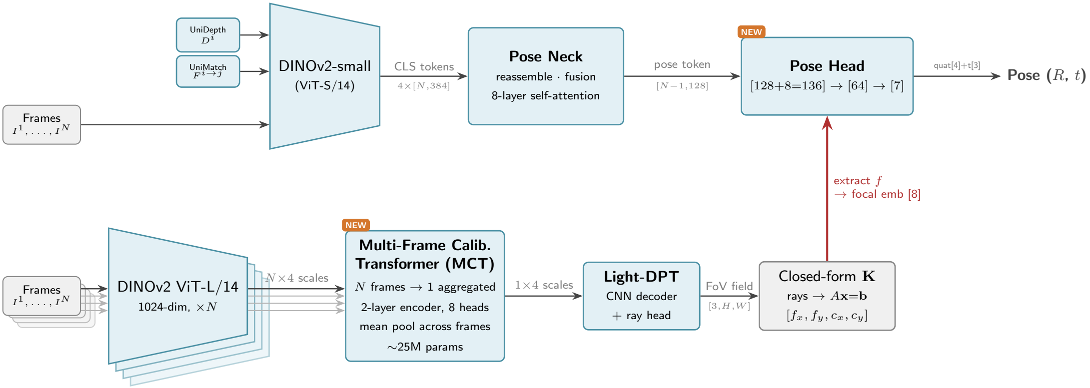


AnyCalib calibration error:

Dataset	AnyCam	AnyCalib
Sintel	45.1%	**30.7%**
TUM-RGBD	63.7%	**11.7%**

- Per-frame only — noisy, inconsistent
- Can we enforce **multi-frame consistency**?
- Can we improve on AnyCalib too?

Our Framework — Fusing Calibration & Pose



Contributions

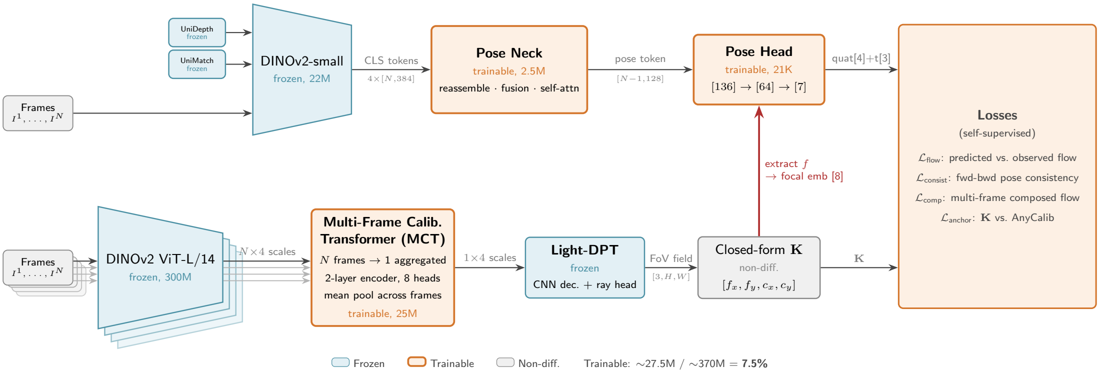
1. Multi-Frame Calibration Transformer (MCT)

- Cross-frame attention on calibration backbone features (4 scales)
- Aggregates N frames $\rightarrow$ 1 consistent calibration
- Architecture-agnostic: inserts between any backbone and decoder
- Only ~ 25M trainable params (7.5% of total)

2. Demonstrated on AnyCam + AnyCalib

- Fuses a self-supervised pose estimator with a calibration network
- Improves *both* pose accuracy and calibration accuracy
- Fully self-supervised — no ground truth labels

Training Setup



Training Losses

Flow Reprojection (self-supervised, main signal):

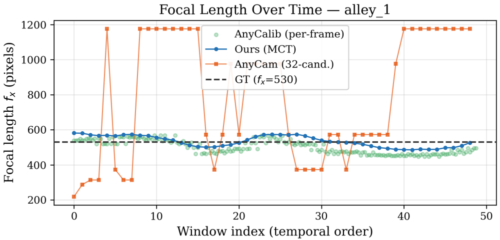
$$\mathbf{x}' = \pi\left(\mathbf{K}\left(R\,\mathbf{K}^{-1}\begin{pmatrix}x\\y\\1\end{pmatrix}d + \mathbf{t}\right)\right), \quad \mathcal{L}_{\text{flow}} = \|\mathbf{x}' - \mathbf{x} - \mathbf{w}_{\text{obs}}\|$$

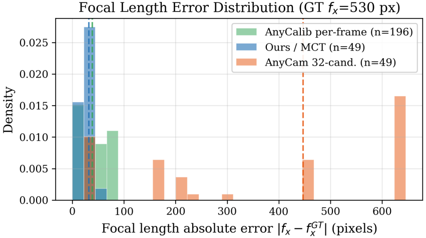
Calibration Anchor — why $\mathbf{K}$ needs constraints:

$$R \approx I,\ \mathbf{t} \approx 0 \quad \Longrightarrow \quad \mathbf{K}^{-1}(d\,\mathbf{K}\mathbf{x} + \mathbf{t}) \approx \mathbf{x} \quad \text{(any } \mathbf{K} \text{ works)}$$

$$\mathcal{L}_{\text{anchor}} = \|\mathbf{K}_{\text{pred}} - \mathbf{K}_{\text{AnyCalib}}\|$$

Results — Calibration Error

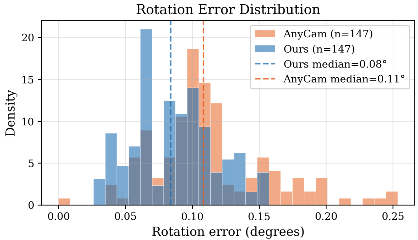


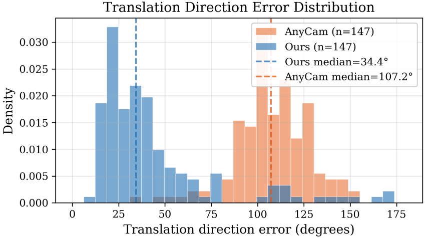


Dataset	Method	MAE (px)↓
Sintel	AnyCam	502.5
	AnyCalib	329.9
	Ours	**300.3**
		−9% vs AnyCalib
TUM-RGBD	AnyCam	234.1
	AnyCalib	43.0
	Ours	**30.3**
		−**32%** vs AnyCalib

$N = 200$ seq. KITTI excluded (OOD cameras).

Results — Pose Error

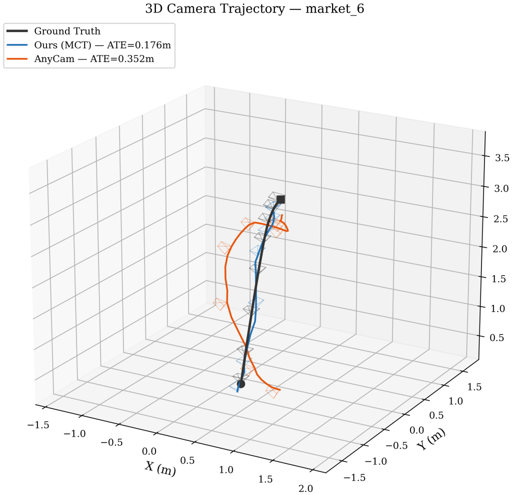




Dataset	Method	Rot. (°)↓ mean / med	Trans. (°)↓ mean / med
Sintel	AnyCam	0.67 / 0.21	89.3 / 86.5
	Ours	**0.60 / 0.19**	**65.0 / 52.3**
		−10% / −8%	−27% / **−40%**
TUM	AnyCam	2.03 / 1.23	93.0 / 97.1
	Ours	**1.39 / 0.71**	**71.9 / 66.9**
		−32% / **−43%**	−23% / −31%
KITTI	AnyCam	0.56 / 0.26	91.1 / 91.5
	Ours	**0.54 / 0.25**	**77.2 / 70.9**
		−2% / −2%	−15% / −22%

$N = 200$ seq. (600 pairs) per dataset. No bundle adjustment.

Trajectory Visualization — Sintel `market_6`



Conclusion & Future Work

Summary:

- Proposed a framework for fusing pretrained calibration & pose models
- MCT: multi-frame calibration consistency via feature-level aggregation
- Demonstrated on AnyCam + AnyCalib: **up to 40% better translation, up to 32% better calibration**

Future work:

- Longer sequences — our model trained at $N = 4$ frames only
- End-to-end training with all components unfrozen, on larger and more diverse data

Thank you! Questions?